1 Fig. 1.1 shows two identical cylindrical metal conductors P and Q, each of length *L* and cross-sectional area *A*.

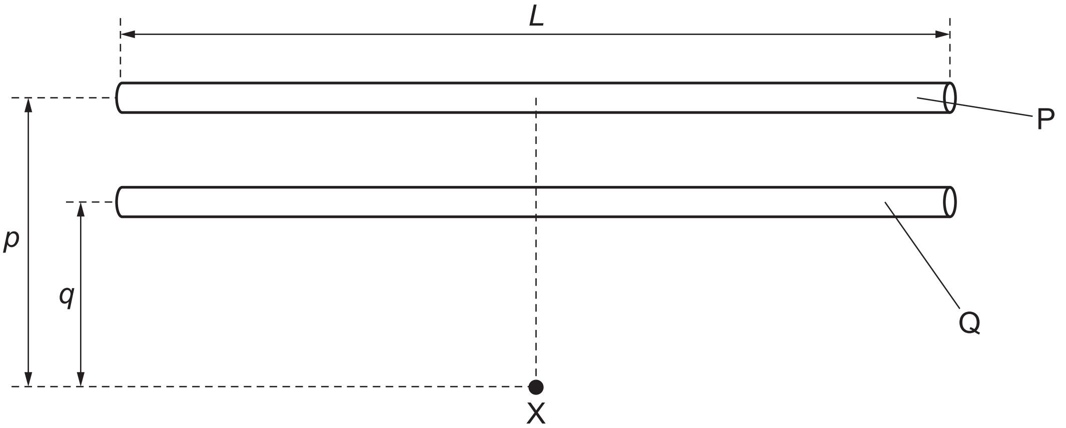

Fig. 1.1

The conductors are placed parallel to each other. The perpendicular distance from the midpoint of P to point X is *p*. The perpendicular distance from the midpoint of Q to point X is *q*.

The two conductors are electrically connected in parallel. This parallel combination is connected in series to a power supply and a resistor. The potential difference *V* between the ends of P is the same as the potential difference between the ends of Q.

The magnetic flux density at X due to the currents in the conductors is *B*.

It is suggested that *B* is related to *p* by the relationship

$$B = \frac{YAV}{Lp} + \frac{YZAV}{Lq}$$

where *Y* and *Z* are constants.

Plan a laboratory experiment to test the relationship between *B* and *p*.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for *Y* and *Z*.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

DO NOT WRITE IN THIS MARGIN

Diagram

....

....

....

....

....

....

....

....

....

....

....

....

....

....

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

© UCLES 2025

[15]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

© UCLES 2025